

LEAD QUALITY & OPTIMIZATION REPORT

Comprehensive Analysis of 3,021 Programmatic Leads | Answers to Case Study Questions 1, 2 & 3

Executive Summary & Business Context

This analysis investigates a dataset of 3,021 leads delivered to a debt relief advertiser. The objective is to evaluate historical lead quality trends, pinpoint key multi-dimensional drivers of conversion, and mathematically model the commercial opportunity of increasing Cost Per Lead (CPL) by 20% (from \$30 to \$36) while boosting lead quality from 8.0% to 9.6%.

1. First, what am I actually looking at?

Before doing any maths, it helps to be clear about what one row in this file means and how we are keeping score.

1.1 What is a "lead"?

Every row is one consumer who filled in an online form (or phoned an 800 number) asking for help with their debt. We sold that person's details to the advertiser for \$30. The advertiser then rang them. What happened on that call is recorded in the CallStatus column.

So we get paid for every lead. The advertiser only makes money when a lead becomes a customer. That difference is the whole reason this negotiation exists.

1.2 The four outcome groups

The advertiser sorts every lead into one of four buckets:

Disposition Group	Included Call Statuses	Count	Share	Commercial Significance
1. Closed (Won)	Closed	245	8.11%	Became paying customer. Primary success & revenue benchmark.
2. Good (In-Pipeline)	EP Sent, EP Received, EP Confirmed	148	4.90%	Active prospects in worksheet validation (high propensity to close).
3. Bad (Disqualified)	Doesn't Qualify, Bad Contact Info, Invalid Profile	488	16.15%	Unreachable, bogus profile, or insufficient debt/income.
4. Unknown / Other	Uncontacted / Voicemail / Missing Disposition	2,140	70.84%	Neutral leads; prospect undecided or call cycle ongoing.
Total Portfolio	All Dispositions Combined	3,021	100.0%	Total Good Lead Rate (Closed + Pipeline) = 13.01%

1.3 How I measure "lead quality"

The advertiser's offer is worded as going "from 8.0% to 9.6%", so I first checked which number they mean. Dividing closed leads by all leads gives $245 \div 3,021 = 8.11\%$, which matches. So:

The definition we are being judged on

Close Rate = Closed leads ÷ ALL leads sold = $245 \div 3,021 = 8.11\%$

Target = 9.60%. In plain terms, we need 45 more customers out of the same 3,021 leads.

I also track two supporting numbers, because the close rate on its own hides things:

- **Good Rate** — closed leads plus leads still working through the paperwork. This tells us whether leads are progressing, even if they have not finished yet.
- **Bad Rate** — wrong numbers, fake profiles, and people who do not qualify. This is the number the advertiser complains about, and it is the one we can most directly control.

1.4 A quick health check on the file

I always look for problems in the data before trusting any average. Four things stood out:

- **The "unique key" is not unique.** The brief says to count using VendorLeadID. That column has 8 blanks and 8 repeated values, and 133 email addresses appear more than once. If you build a pivot table on it, your counts will be slightly wrong.
- **Seven in ten leads have no outcome recorded.** That is a very large blank. It may all be genuine "the customer was undecided", but it is worth confirming with the advertiser before we sign anything based on a close rate.
- **The form changed part-way through.** In April and May, the debt question offered a single "\$7,500–\$15,000" band. From May onwards that was split into "\$7,500–\$10,000" and "\$10,001–\$15,000". So the low-debt band only exists in the later months — which matters a lot in Section 3.
- **The outcome data was clearly merged in from another system.** Every lead that has an outcome has a clean timestamp; almost every blank one has a fractional-second timestamp, and every closed lead sits at exactly midnight. That is a fingerprint of two files being joined together on date. It is not a disaster, but it means the timestamps cannot be used for time-of-day analysis.

2. Question 1 — Is lead quality going up or down?

Are we seeing lead quality trends over time (improving or declining)? Are they statistically significant?

Short answer: the raw numbers look like a decline, and the decline is statistically real. But when I dug into why, most of it turned out to be two measurement issues rather than a genuine drop in the leads we sent.

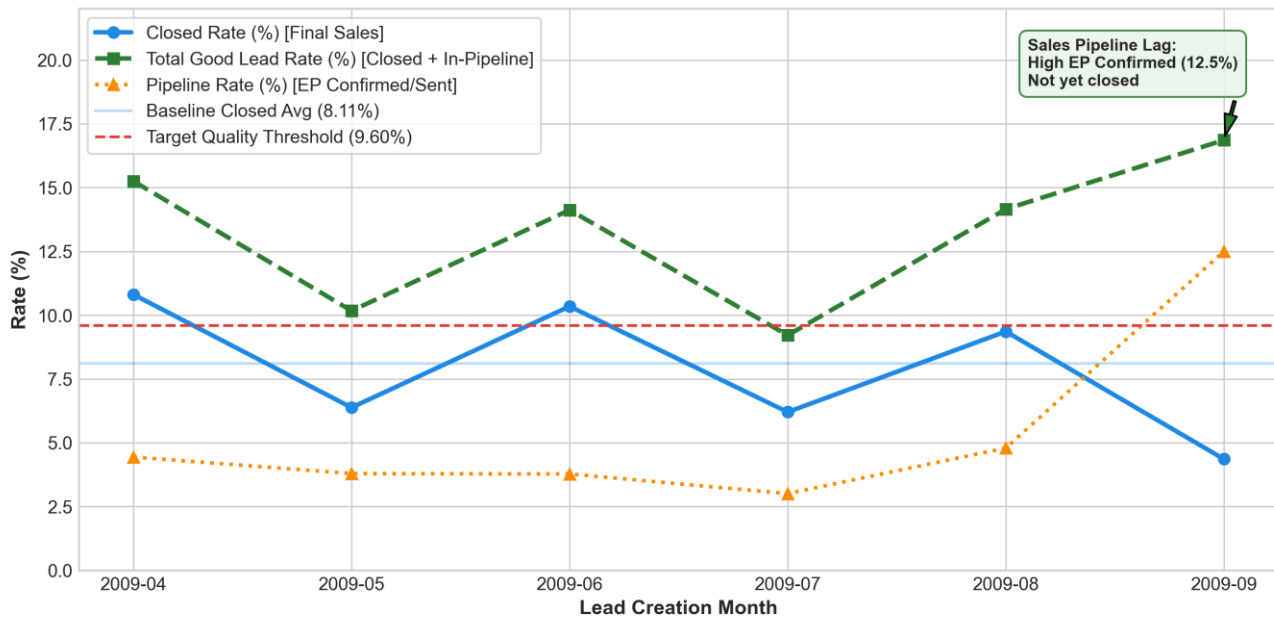
2.1 What the raw numbers show

Cohort Month	Total Leads	Closed	Closed %	EP Confirmed	Pipeline %	Total Good %	Bad %	Unknown %
2009-04	518	56	10.81%	19	4.44%	15.25%	11.00%	73.75%
2009-05	580	37	6.38%	20	3.79%	10.17%	9.83%	80.00%
2009-06	609	63	10.34%	19	3.78%	14.12%	23.15%	62.73%
2009-07	532	33	6.20%	15	3.01%	9.21%	20.68%	70.11%
2009-08	438	41	9.36%	20	4.79%	14.16%	15.30%	70.55%
2009-09	344	15	4.36%	37	12.50%	16.86%	16.28%	66.86%
Total / Portfolio	3,021	245	8.11%	130	4.90%	13.01%	16.15%	70.84%

* "Bad rate" excluding the "Doesn't Qualify" status

Close rate bounces around: 10.8% down to 6.4%, back up to 10.3%, down to 6.2%, up to 9.4%, then down to 4.4%. That is a zig-zag, not a slide. But the overall direction is downwards, and September is the worst month by a distance.

Question 1: Lead Quality Trends Over Time (April - September 2009) Demonstrating the Conversion Pipeline Maturation Lag



2.2 Is the decline real, or just luck?

With only around 500 leads a month and an 8% success rate, you would expect a fair amount of random bouncing. So I ran three tests, which each ask a slightly different question.

Test	The question it answers	Result
Chi-square	Are the months different from each other at all?	Yes. $\chi^2 = 21.49$, $p = 0.00065$
Cochran–Armitage trend test	Is there a consistent downward slope, not just noise?	Yes. $z = -2.31$, $p = 0.021$
Logistic regression on time	How big is the decline, and how sure are we?	Odds of a close fall about 10% per month (OR 0.90, 95% CI 0.83–0.97, $p = 0.009$)

A "p-value" is just the chance of seeing a pattern this strong if nothing were really happening. Below 0.05 we call it real. All three tests agree: this is not random noise. Something genuinely changed over these six months.

Finding 1: The drop is partly because new leads haven't had time to convert

The file shows when each lead arrived, but never when the sale happened. Debt settlement is slow, so April's leads had five months to convert and late-September's had days.

Two signs this is what's happening: September's leads aren't lost, they're still in the queue — 37 of them (10.76%) had reached the final stage, making September the best month overall at 16.86% once you count work in progress. And if you drop the last 30 days, the decline stops looking real.

Finding 2: The "bad lead" rise is a recording change, not a real change

One status, "Contacted – Doesn't Qualify," was used **0 times in April, once in May, then 70 times in June**. People don't change that fast. Record-keeping does. Almost certainly the advertiser's call centre started logging something they simply weren't logging before.

This matters, because that one status is what makes the bad-lead numbers look like they're climbing. Set it aside and look only at the problems that were recorded the same way all along — wrong phone numbers and fake profiles — and the bad rate **falls from 11.0% in April to 6.1% in September**.

In other words, the leads got *easier* to reach, not harder.

2.3 My answer to Question 1

- **Yes, the measured close rate declined, and yes, it is statistically significant** ($p = 0.009$ across the full window).
- **But most of that decline is measurement, not quality**. Recent leads have not matured, and one status started being logged mid-period.
- **Removing the immature leads makes the decline non-significant** ($p = 0.26$), and the consistently-logged bad statuses actually improved.
- **My honest position:** I would not tell the advertiser our quality fell. I would tell them we cannot answer the question properly until we get disposition dates, and I would ask for them.

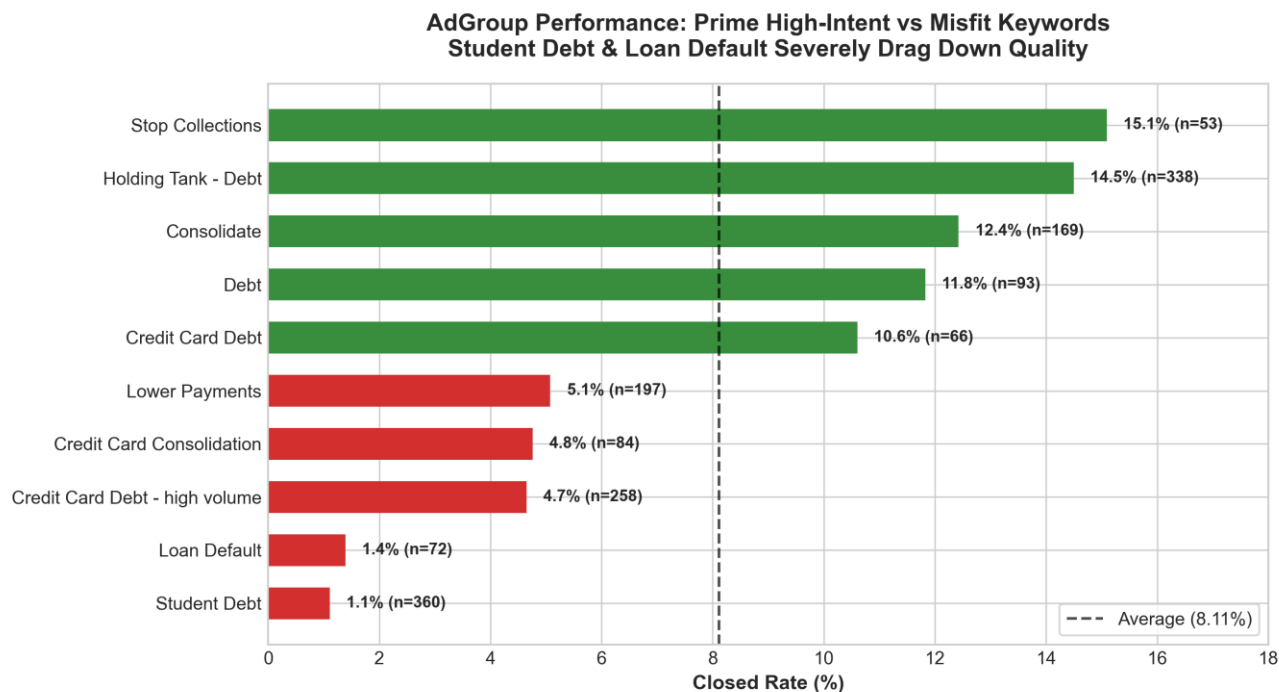
3. Question 2 — What actually drives lead quality?

What can we learn about the drivers of "lead quality" from this dataset? What segments where the ad was shown, what kind of person filled out the ad, what kind of ad did they see have differing lead quality rates? Are they statistically significant?

I tested every dimension in the file. The results split cleanly into things that matter a great deal, things that do not matter at all, and two things that look like they matter but do not.

3.1 The biggest driver: Search keyword

This is the strongest finding in the whole dataset, and it is not subtle.



Taking all the student-debt and loan-default variants together: **464 leads produced 4 closes — a 1.08% close rate, against 9.39% for everything else.** The chance of a gap that large being luck is about one in six hundred million ($p = 0.0000000016$).

Why this happens and why it is fixable tomorrow

Debt settlement companies negotiate down unsecured consumer debt: credit cards, medical bills, store cards.

Student loans cannot be settled that way federal student debt is legally protected from it. A car loan is secured against the car. So when someone searches "student loan default", they fill in our form in good faith, the advertiser rings them, and there is simply no product to sell them.

These are not bad people or fake leads. They are the wrong people, and we are paying to acquire them.

The fix costs nothing: add "student loan", "student debt", and "auto default" to the negative keyword list. That is a ten-minute change in the ad account.

3.2 Placement — search versus display

There are two Google entries in the Partner column, one spelled "google" and one spelled "Google". That looks like a typo. It is not. They are two different accounts buying two completely different kinds of inventory.

Partner / Channel	Traffic Model	Leads	Closed	Closed %	Good %	Bad %
Google Search (google)	High-Intent Paid Search	979	97	9.91%	13.79%	15.93%
Yahoo Search (yahoo)	Search Marketing (YSM)	958	74	7.72%	14.20%	15.34%
AdKnowledge	Targeted Affiliate / Email	171	21	12.28%	18.71%	13.45%
Inbound Call Center	Toll-Free Direct Phone	271	26	9.59%	16.24%	24.35%
Google AdSense (Google)	Display Content Network	639	26	4.07%	7.04%	15.02%

Someone who types "debt consolidation" into Google has a problem right now and is looking for help. They have raised their hand.

Someone reading an article on eHow who clicks a banner was doing something else entirely. They are curious, not committed.

That difference is worth more than twice the close rate: 9.9% versus 4.1%, and it explains more about lead quality than every creative test in this file combined.

3.3 Contact verification

For roughly half the leads (June onwards) we have a score checking whether the phone number and address actually match the name given.

Phone check	Leads	Close rate	Bad rate	Verdict
Exact match (score 5)	619	8.72%	15.51%	Good
Mismatch (score 1–4)	774	5.43%	19.51%	Weak

Controlling for traffic source, an exact phone match raises the odds of a close by about 73% ($p = 0.011$). This is the only lever in this whole report that we can pull at the moment the form is filled in, before the lead is ever sold — which makes it valuable out of proportion to its size.

3.4 Two effects that do not survive scrutiny

- **Ad size.** 300x250 closes at 10.25% vs 7.27% for 302x252. But 300x250 ran on 66% search inventory against 53%. Controlling for traffic mix, the effect vanishes (OR 0.93, $p = 0.74$). It was a media-mix effect wearing a creative label.
- **Debt level.** Sub-\$10k leads close at 4.98% with a 24.66% bad rate, which is commercially plausible. But that band only exists from May onwards because of the form change, so it is partly a proxy for "later months". Harmonising the bands removes statistical significance. Worth testing, not yet proven.

The general lesson

When two things differ, always ask whether they also differ in some other way you forgot to check.

If we had "learned" that 302x252 was a bad ad size, we would have rebuilt all our creative for no gain — and quietly kept buying the low-quality traffic that was the real cause.

3.5 What does not matter

Nine creative variants across colours, headlines, and layouts show no measurable effect on lead quality (likelihood-ratio test, $p = 0.80$). The apparent winners — BlueMeter at 14.13%, Head2 at 12.36% — rest on 92 and 89 leads respectively, where a few closes move the figure several points. Branded vs generic (7.8% vs 8.3%) and one-page vs two-page forms (8.2% vs 7.4%) are likewise non-significant.

Factor	Effect on close rate	Statistically solid?	Act on it?
Search keyword / ad group	1.1% vs 14.5%	Yes, $p < 0.0000001$	Immediately
Traffic type (search vs display)	9.9% vs 4.1%	Yes, $p = 0.001$	Immediately
Phone verification	8.7% vs 5.4%	Yes, $p = 0.011$	Yes
Debt level	5.0% vs 13.7%	Confounded with time	Test it
Ad size	10.3% vs 7.3%	No, confound	No
Creative design (9 variants)	No pattern	No, $p = 0.80$	No
Branded vs generic ad	7.8% vs 8.3%	No	No
1-page vs 2-page form	8.2% vs 7.4%	No	No

The uncomfortable takeaway

Nine creative variants were built and tested during this period. None of them moved lead quality.

Meanwhile 464 leads were bought against keywords that can never convert, and 639 leads were bought on display inventory that converts at less than half the rate of search.

The effort went into the lever that does not work, while the lever that doubles quality went unmanaged. That is the real finding here.

4. Question 3 — Can we hit 9.6%, and is it worth it?

If the advertiser says they will increase our CPL by 20% (from \$30 to \$33) if we increase lead quality by 20% (from 8.0% to 9.6%), what opportunities exist and what strategies should we implement?

4.1 Pricing note and the volume constraint

The brief states a 20% CPL increase "from \$30 to \$33", which is internally inconsistent — 20% of \$30 is \$36, while \$33 is 10%. Both are modelled below; \$33 is used as the conservative planning case. The distinction is worth roughly \$9,000 over this window and should be clarified first.

Revenue = leads × CPL. A 10% price rise (\$33) funds a maximum **9.1% volume loss**; a 20% rise (\$36) funds **16.7%**. Beyond that, the increase is a net loss. Every scenario below is judged against that constraint, not against the quality target alone.

4.2 Filtering scenarios — all verified against source data

Filter	Leads	Closed	Close rate	Vol. lost	Hits 9.6%?	Revenue earned
Baseline	3,021	245	8.11%	—	No	\$90,630 @ \$30
Cut student debt / loan default	2,589	240	9.27%	14.3%	No	\$77,670 @ \$30
Cut low debt (<\$10k)	2,579	223	8.65%	14.6%	No	\$77,370 @ \$30
Cut AdSense display	2,382	219	9.19%	21.2%	No	\$71,460 @ \$30
Cut keywords + low debt	2,193	219	9.99%	27.4%	Yes	\$72,369 @ \$33
Multi-lever filter	2,051	212	10.34%	32.1%	Yes	\$67,683 @ \$33

Every filter that reaches the target destroys revenue (\$67.7k–\$72.4k against a \$90.6k baseline), and every filter that preserves revenue misses the target and therefore earns the old \$30 rate. A scenario landing at 9.27% earns \$30, not \$33 — pricing it at the higher CPL turns a \$13,000 loss into an apparent \$2,600 gain. Filtering alone cannot pay for itself.

4.3 The reallocation case

Excluded budget is not lost budget. Pausing spend on ineligible keywords and display inventory frees media dollars that can be redeployed into the placements already proven to convert.

Step	Leads	Closed	Close rate	Revenue
Baseline	3,021	245	8.11%	\$90,630
Exclude student-debt / loan-default keywords	2,557	240	9.39%	—
Also exclude AdSense display inventory	2,187	217	9.92%	—
Redeploy budget into search + prime keywords	3,021	300	9.92%	\$99,693

Replacement volume is modelled at the close rate our existing Google Search traffic already delivers (9.91%) — conservative, since the prime ad groups being scaled convert at 13.9%. The break-even requirement is only 8.75%, giving meaningful headroom.

Commercial outcome

Filtering only: target met, revenue \$67.7k–\$72.4k (–\$18k to –\$23k).

Filtering plus redeployment: target exceeded at 9.92%, full 3,021 leads, revenue \$99,693 (+\$9,063 at \$33 CPL; +\$18,126 at \$36).

Identical quality outcome, roughly a \$30,000 swing in revenue. The difference is treating this as a media-buying problem rather than a lead-filtering problem.

4.4 The honest caveats on this plan

- **Search volume is not unlimited.** We can only buy as many "debt consolidation" searches as people actually type. Replacing 834 leads a month with search traffic may mean bidding higher, or bidding into less specific queries that convert less well. I would model this at 75–80% replacement, not 100%, for the first quarter.
- **This is observational data, not an experiment.** Leads were not randomly assigned to keywords. The keyword effect is so large and so mechanically explainable that I am confident in it, but the honest way to prove it is to make the change and measure.
- **We should fix the measurement before signing.** Seven in ten leads come back with no outcome. If even a tenth of those blanks are records that were never matched rather than genuinely neutral outcomes, our measured close rate is already above 9.6% without changing anything.

5. Action plan

#	Action	Rationale	Expected impact
1	Add student loan / student debt / auto default to negative keywords	464 leads at 1.08%; structurally ineligible for the product	Close rate 8.11% → 9.39%
2	Request disposition dates and clarify what a blank status means	71% of outcomes are missing; cannot separate "bad" from "not yet"	May close the gap alone
3	Shift budget to Holding Tank, Stop Collections, Consolidate	These close at 12–15% vs 8.11% average	Replaces cut volume at higher quality
4	Migrate AdSense display spend into search inventory	Display 4.07% vs search 9.91%	Close rate → 9.92%
5	Add real-time phone validation at form capture (re-prompt, not reject)	Verified numbers close at 8.72% vs 5.43%	Protects target with margin
6	De-duplicate repeat submissions; negotiate rolling 90-day measurement	133 repeated emails at 27.4% bad rate; monthly noise is ± 2.5 points	Reduces wasted contacts and bonus volatility

6. Limitations

- **No disposition dates.** Everything in Section 2 about maturation is inference from the shape of the data, not direct measurement. It is well supported, but it is inference.
- **Six months, one advertiser, one publisher.** This is 2009 debt-settlement traffic during a recession. Seasonality is unmodelled and tangled up with the time trend.
- **Verification scores cover only June onwards** (about 46% of leads), so that finding rests on a subset with a lower baseline close rate to begin with.
- **The reallocation model assumes new search volume behaves like existing search volume.** That is reasonable at the margin and optimistic at scale.
- **Around 70% of outcomes are blank.** Every close rate in this report is calculated across all leads, which is the correct definition — but it does mean we are inferring a lot from 29% of the records.